Predictive Uncertainty in Deep Medical Image Denoising: A Bayesian Approach

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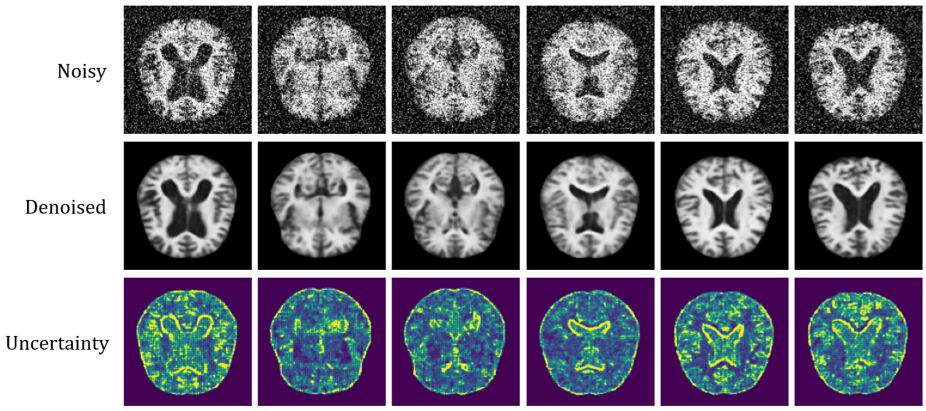

Figure 1: Our model can estimate the uncertainty corresponding to the main objective of denoising MRI brain scan images.

ABSTRACT

Deep learning methods, such as autoencoders, have emerged as powerful tools for denoising medical images, yet these models often lack interpretability and do not provide uncertainty quantification in generating these images. In this paper, we integrate Monte Carlo dropout to approximate Bayesian neural networks in denoising autoencoders, with the goal of quantifying model uncertainty in generating denoised MRI brain scan images at the pixel level. Results show that this integration provides reasonable estimates of model uncertainty, but comes at the expense of predictive accuracy for more complex datasets. Particularly for inaccurately predicted images, the model identifies specific regions with higher uncertainty. Furthermore, we observe a positive correlation between model-estimated uncertainty and prediction error. By providing uncertainty estimates, this study contributes to well-informed decision-making in the medical field where accurate image denoising is crucial[1].

KEYWORDS

Image denoising, Denoising autoencoders, Model uncertainty, Monte Carlo Dropout, Bayesian neural networks, deep learning

[1]The content of this thesis is the sole responsibility of the author and does not reflect the view of Erasmus School of Economics or Erasmus University.

1 INTRODUCTION

Image denoising plays a pivotal role in fields where high-quality images are crucial for decisionmaking, such as medical imaging, autonomous driving, and remote sensing [Abdar et al. 2021]. In recent years, deep learning techniques, such as autoencoders, have emerged as powerful tools in the field of image denoising. Among these, denoising autoencoders (DAEs) are specifically designed to remove noise from images [Vincent et al. 2008]. Despite their impressive performance, these models are still regarded as black boxes and an area that remains relatively unexplored is the estimation of uncertainty in their predictions.

In medical imaging, introducing uncertainty estimates, as displayed in Figure 1, could make decision-making more informed. For instance, Kascenas et al. [2023] have shown that DAEs are effective for tumor identification in MRI brain scans. In such applications, uncertainty quantification in these (denoised) images can provide medical professionals with insights into the quality of the images, potentially reducing the risk of misdiagnosis [Zou et al. 2023]. In these contexts, traditional deterministic neural networks, including standard (denoising) autoencoders, do not naturally provide a measure of uncertainty [Gal and Ghahramani 2016].

This paper aims to integrate Monte Carlo (MC) dropout, a technique for approximating Bayesian inference in neural

networks, with denoising autoencoders to quantify model uncertainty within image denoising predictions of MRI brain scan images at the pixel level. Specifically, we apply MC dropout only to the (generating) decoder of the autoencoder network. The study is guided by two central research questions: (1) To what degree can Monte Carlo dropout, as applied to denoising autoencoders, effectively quantify model uncertainty in the prediction of denoised MRI brain scan images? and (2) What impact does the integration of uncertainty quantification through Monte Carlo dropout have on the accuracy of denoising autoencoders on these images?

The primary objective is to develop an image denoising framework that not only preserves image quality but also provides a quantifiable measure of uncertainty in its predictions. Additionally, the study investigates the quality of the uncertainty predictions and explores whether denoising accuracy is preserved in MRI brain scan images. Throughout this paper, we refer to the integration of MC Dropout in denoising autoencoders as Bayesian Denoising Autoencoders (BDAEs).

Contribution: This study broadens the scope of current research on deep denoising models in medical imaging, which is traditionally centered on improving denoising accuracy. This paper emphasizes the interpretability of these models by providing uncertainty estimates at the pixel level. Our findings demonstrate that BDAEs provide qualitatively reasonable estimates of model uncertainty, but this benefit comes at the expense of losing predictive accuracy for more complex datasets. In instances where the model encounters difficulty in accurately denoising the images, it identifies specific regions with higher uncertainty. Moreover, we observe a moderate positive correlation between the uncertainty estimated by the model and the actual prediction error. By providing uncertainty estimates, the study contributes to making decision-making more informed in the medical field, where accurate image denoising is crucial.

The paper is structured as follows: in Section 2, we provide an overview of previous work in image denoising and the role of uncertainty quantification. Section 3 outlines the methodology. The datasets are presented in Section 4. Then, the results of our research and its implications are presented in Section 5. Finally, Section 6 states the conclusions and potential future work.

2 RELATED WORK

Our work lies in the intersection of two strands of research: estimating uncertainty in deep neural networks and image denoising models. This section provides an overview of existing methods, beginning with image denoising models and then transitioning to uncertainty quantification techniques.

Over the years, various deep learning techniques have significantly advanced image denoising due to their ability to learn complex mappings between noisy and clean images [Tian et al. 2020]. Recent successful (complex) approaches include convolutional neural networks (CNNs), with architectures like U-Net and DnCNN, and generative adversarial networks (GANs) [Zhong et al. 2020]. We focus on denoising autoencoders (DAEs) [Vincent et al. 2008] due to their simplicity and computational efficiency, while still offering strong performance. Initially used to learn compact representations of data, DAEs have also proven effective in image denoising due to their noise-removal capabilities. For instance, Kascenas et al. [2023] demonstrated the effectiveness of DAEs in identifying tumors in brain MRI images, and Gondara [2016] showed that convolutional DAEs can reduce noise in X-ray images. Despite their success, DAEs often lack insights into uncertainty, as the complex nature of these models makes it difficult to understand their decision-making processes.

In deep learning, common approaches for uncertainty quantification (UQ) in neural networks include ensemble techniques [Lakshminarayanan et al. 2017] and Bayesian inference [Gal et al. 2016]. Our approach, using MC dropout, falls under the latter category, and is chosen for its simplicity and computational efficiency compared to other methods such as variational inference and Markov Chain Monte Carlo [Abdar et al. 2021].

UQ in image denoising models is a topic that has not received much attention in the literature. Ma et al. [2021] use ensemble techniques with decoupled attention paths to estimate model uncertainty, but this approach is complex and computationally intensive. In contrast, our work implements MC dropout on a DAE network, offering a simpler approach compared to deep ensembles. The work most closely related to ours is by Laves et al. [2020], which uses a deep image prior (iteratively training a randomly initialized network on a single image) for denoising medical images and extends this using MC dropout for uncertainty estimation. The main difference in our approach is that we train the parameters on a dataset that includes multiple pairs of noisy and clean images and use this for image denoising, which Jo et al. [2021] demonstrated to produce better denoising results than deep image priors.

Other approaches for quantifying uncertainty in autoencoders (excluding DAEs) involve Bayesian autoencoders [Yong and Brintrup 2022] and variational autoencoders [Saetta et al. 2024]. These methods aim to quantify uncertainty in anomaly detection scenarios by generating stochastic predictions through the sampling of latent variables. While uncertainty in the latent representation is valuable for anomaly detection, our work employs DAEs for image denoising, focusing on uncertainty in the output.

3 METHODOLOGY

This section covers the methods and models used in the paper. First, traditional autoencoders are introduced. Following this, Bayesian neural networks and Monte Carlo dropout are discussed. Finally, the Bayesian denoising autoencoder is presented, our novel contribution to the literature.

3.1 The Denoising Autoencoder

A denoising autoencoder (DAE) is a modified version of the traditional autoencoder [Bengio and LeCun 2007] that uses a corrupted, partially destroyed data point $\tilde{\mathbf{x}}$ as input, and is trained to reconstruct the original "clean" input [Vincent et al. 2008]. The corrupted input is obtained by partially destroying initial input $\mathbf{x}$ to get a corrupted version $\tilde{\mathbf{x}}$, by using a stochastic mapping $\tilde{\mathbf{x}} \sim q_{\mathcal{D}}(\tilde{\mathbf{x}}|\mathbf{x})$. In this paper, *Gaussian noise* is used as corruption process, as Gravel et al. [2004] have shown the prevalence of Gaussian noise in medical images arising from electronic components. The DAE network architecture consists of an inference model, the encoder, and a generative model, the decoder. The objective is to use the encoder to compress corrupted datapoint $\tilde{\mathbf{x}} \in \mathbb{R}^n$ into a hidden representation vector $\mathbf{z} \in \mathbb{R}^m$, where $m < n$, through a deterministic (neural network) mapping $\mathbf{z} = f_\theta(\tilde{\mathbf{x}})$ [Vincent et al. 2008]. In the case of a simple one-layer encoder this mapping is represented as $f_\theta(\mathbf{x}) = s(\mathbf{W}\mathbf{x} + \mathbf{b})$, with parameters $\theta = \{\mathbf{W}, \mathbf{b}\}$ and activation function s. This latent representation vector $\mathbf{z}$ is then used by the decoder to attempt to reconstruct the original "clean" data, creating "reconstruction" vector $\mathbf{y} = g_{\theta'}(\mathbf{z}) \in \mathbb{R}^n$. This results in every input $\mathbf{x}^{(i)}$ during training being mapped to a corresponding hidden representation $\mathbf{z}^{(i)}$ and a reconstructed output $\mathbf{y}^{(i)}$. Initially, autoencoders typically used simple one-layer encoders and decoders. However, modern approaches often use deep multi-layer neural networks to improve performance, with convolutional networks being used for images [Berahmand et al. 2024]. The architecture of the DAE is illustrated in Figure 2.

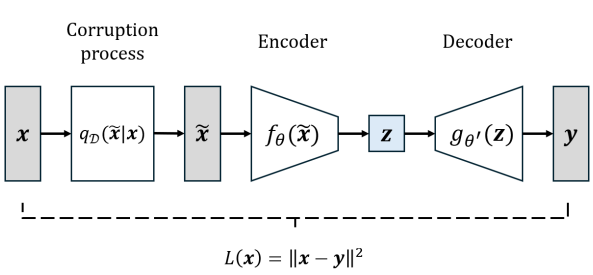

Figure 2: Architecture and objective of a denoising autoencoder.

Note that in the case of grayscale image data, pixel values are normalized to fall within the range of 0 to 1, such that $\mathbf{x}^{(i)}, \mathbf{y}^{(i)} \in [0, 1]^n$. The parameters of the model are adjusted to minimize the *average reconstruction error*:

$$\theta^*, \theta'^* = \arg\min_{\theta,\theta'} \frac{1}{n} \sum_{i=1}^{n} L\left(\mathbf{x}^{(i)}, \mathbf{y}^{(i)}\right) \tag{1}$$

$$= \arg\min_{\theta,\theta'} \frac{1}{n} \sum_{i=1}^{n} L\left(\mathbf{x}^{(i)}, g_{\theta'}\left(f_\theta(\mathbf{x}^{(i)})\right)\right), \tag{2}$$

where L is a loss function. In the case of image data, this loss function is typically the *squared error*, i.e. $L(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|^2$. As in traditional autoencoders, the primary purpose of DAEs is to learn robust representations of the data [Bengio and LeCun 2007]. In contrast, DAEs intentionally corrupt the input data to force the model to learn more robust features. This makes them more suitable for applications where data can be noisy or incomplete, such as in image denoising.

Gaussian Noise In this paper, we consider Gaussian noise, parametrized by the desired "destruction" variance σ^2 [Liu and Lin 2012]. Under the Gaussian noise scheme, for every input image $\mathbf{x}$, each element (or pixel) is corrupted by adding a random variable drawn from a Gaussian distribution with zero mean and variance σ^2. Mathematically, let $\mathbf{x}$ represent the original input data (e.g. a vector of pixel values), then the corrupted input $\tilde{\mathbf{x}}$ can be defined as

$$\tilde{\mathbf{x}} = \mathbf{x} + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, \sigma^2\mathbf{I}), \tag{3}$$

where ε represents the Gaussian noise added to each element of $\mathbf{x}$, and σ controls the amount of noise added.

3.2 Bayesian Deep Learning

Bayesian deep learning offers a principled framework for quantifying uncertainty in neural networks [Gal and Ghahramani 2016]. Unlike traditional neural networks that output deterministic predictions, Bayesian neural networks provide a probabilistic approach, allowing us to estimate uncertainty associated with predictions.

Model Uncertainty An ideal neural network for decision-making should address all sources of uncertainty. There are two primary types of uncertainty in a model: data (or aleatoric) uncertainty and model (or epistemic) uncertainty. Data uncertainty represents irreducible uncertainty that remains present even with unlimited data availability, such as variability in image data caused by lighting conditions. On the other hand, model uncertainty captures the uncertainty within the model and its parameters [Sensoy et al. 2018]. This type of uncertainty arises from a lack of knowledge about the model, and it decreases as more data becomes available. The total uncertainty can be decomposed into these two types of uncertainty (for regression) using the law of total variation

[Malinin et al. 2020]:

$$\underbrace{\mathbb{V}_{p(\mathbf{y}|\mathbf{x},\mathcal{D})}[\mathbf{y}]}_{\text{Total Uncertainty}} = \underbrace{\mathbb{V}_{p(\Theta|\mathcal{D})}\left[\mathbb{E}_{p(\mathbf{y}|\mathbf{x},\Theta)}[\mathbf{y}]\right]}_{\text{Model Uncertainty}} + \underbrace{\mathbb{E}_{p(\Theta|\mathcal{D})}\left[\mathbb{V}_{p(\mathbf{y}|\mathbf{x},\Theta)}[\mathbf{y}]\right]}_{\text{Expected Data Uncertainty}}.$$

In this paper, we use Bayesian deep learning techniques to quantify *model uncertainty* as it captures the model's confidence in its predictions, enabling us to identify when the model is unsure about its output, given the input [Abdar et al. 2021].

Probabilistic Framework To quantify model uncertainty in neural network predictions, we transition from a deterministic to a (probabilistic) Bayesian approach, which incorporates uncertainty into both the model and its predictions. In traditional neural networks, the unknown parameters are determined using a backpropagation algorithm, which produces a single point estimate for the model weights and the predicted value. Bayesian approaches provide a natural probabilistic representation of uncertainty by learning from the data and incorporating this uncertainty into the predicted values. The concept of Bayesian neural networks involves treating the model weights as random variables, each with its own probability distribution, to determine the probability distribution of the predicted values [Neal 2012].

Bayesian Neural Networks We define a Bayesian neural network by assigning a prior distribution to the model weights. When new data is introduced, this prior distribution is updated with the new information to form the posterior distribution [Gal and Ghahramani 2016]. The model learns from new data using Bayes' Rule. Here, the model parameters θ are treated as random variables, and a prior distribution $P(\theta)$ is assigned to them. The posterior distribution $P(\theta|\mathcal{D})$ is then computed using Bayes' rule:

$$P(\theta|\mathcal{D}) = \frac{P(\mathcal{D}|\theta)P(\theta)}{P(\mathcal{D})}, \tag{4}$$

where $\mathcal{D} = \{\mathbf{x}^{(i)}, \mathbf{y}^{(i)}\}_{i=1}^{N}$ represents the training set. Samples from the posterior distribution should produce explanations that align with the observations in the training data. However, for data that deviates significantly from $\mathcal{D}$, each set of parameters may yield different predictions. Consequently, estimates of the model uncertainty can be derived by analyzing the variability in these predictions.

To include the parameter uncertainty in the predicted output $\mathbf{y}^{(i)}$ for the unseen data $\mathbf{x}^{(i)}$, we calculate the predictive distribution by integrating over all possible model configurations for θ:

$$P(\mathbf{y}^{(i)}|\mathbf{x}^{(i)}, \mathcal{D}) = \int P(\mathbf{y}^{(i)}|\mathbf{x}^{(i)}, \theta)P(\theta|\mathcal{D})d\theta. \tag{5}$$

This integral is often intractable, and therefore to compute the posterior distribution, approximations or numerical methods are used.

3.3 Monte Carlo Dropout

One common method to approximate the posterior distribution in Equation (5) is by using Monte Carlo (MC) Dropout. It is able to estimate model uncertainty efficiently without requiring significant modifications to standard neural networks [Gal and Ghahramani 2016]. The method leverages dropout, a regularization technique commonly used to prevent overfitting, and repurposes it to estimate the uncertainty in model predictions.

Dropout Regularization Dropout is traditionally used during training as a regularization technique. The idea is to randomly set a subset of activations to zero with a fixed independent probability p known as the dropout rate, at each training step. This forces the network to learn robust representations. Formally, given a neural network layer $\mathbf{a}$, dropout is applied by element-wise multiplying $\mathbf{a}$ with a binary mask $\mathbf{M}$, sampled from a Bernoulli distribution $\mathcal{B}(p)$:

$$\mathbf{a}_{\text{dropout}} = \mathbf{a} \odot \mathbf{M}, \qquad \mathbf{M} \sim \mathcal{B}(p), \tag{6}$$

where p is the dropout rate and $\odot$ denotes the element-wise multiplication. During training, this process effectively trains an ensemble of sub-networks, as each forward pass through the network can be seen as training a different thinned version of the network. However, during the inference stage, dropout is typically deactivated, and all neurons are utilized for making predictions.

Uncertainty Estimation In MC dropout, Bayesian inference is approximated by applying dropout not only during training but also during inference. By enabling dropout during inference and performing multiple stochastic forward passes, the network produces a distribution of predictions for each input, which approximates the posterior distribution over the model's parameters [Gal and Ghahramani 2016]. More formally, consider a neural network trained with dropout regularization and let $\mathbf{x}$ be the input data, and θ the network parameters. During inference, dropout is applied, resulting in a set of stochastic predictions. Each forward pass $t = 1, ..., T$ through the network with dropout results in a prediction $\hat{\mathbf{y}}^{(t)}$. To approximate the posterior distribution over the model parameters θ, dropout masks $\mathbf{M}^{(t)}$ are randomly sampled, zeroing out neurons in the network with probability p. The network's output for stochastic forward pass t is computed as

$$\hat{\mathbf{y}}^{(t)} = f_\theta(\mathbf{x}; \mathbf{M}^{(t)}), \qquad \mathbf{M}^{(t)} \sim \mathcal{B}(p), \tag{7}$$

where $\mathbf{M}^{(t)}$ represents the dropout mask applied during the t-th forward pass. The final prediction for the input $\mathbf{x}$ is obtained by averaging the predictions from the T stochastic

forward passes, which approximates the mean of the posterior distribution $p(\mathbf{y}|\mathbf{x}, \mathcal{D})$:

$$\hat{\mathbf{y}} = \mathbb{E}[\mathbf{y}|\mathbf{x}, \mathcal{D}] \approx \frac{1}{T} \sum_{t=1}^{T} \hat{\mathbf{y}}^{(t)}. \tag{8}$$

The model uncertainty is captured by the (sample) variance of its predictions. The sample variance $\hat{\sigma}^2$ of the predictive distribution can be estimated as

$$\hat{\sigma}^2 = \mathbb{V}[\mathbf{y}|\mathbf{x}, \mathcal{D}] \approx \frac{1}{T-1} \sum_{t=1}^{T} (\hat{\mathbf{y}}^{(t)} - \hat{\mathbf{y}})^2. \tag{9}$$

Bayesian Interpretation From a Bayesian perspective, MC Dropout approximates the posterior distribution $p(\mathbf{y}|\mathbf{x}, \mathcal{D})$, by averaging over the dropout masks:

$$P(\mathbf{y}|\mathbf{x}, \mathcal{D}) \approx \frac{1}{T} \sum_{t=1}^{T} P(\mathbf{y}|\mathbf{x}, \theta, \mathbf{M}^{(t)}). \tag{10}$$

Here, each dropout mask $\mathbf{M}^{(t)}$ can be seen as a sample from an approximate posterior distribution over the network's weights. This interpretation aligns with the variational Bayesian framework, where dropout acts as a form of variational inference that approximates the posterior by introducing randomness during both training and inference. By performing multiple stochastic forward passes, MC dropout samples from this approximate posterior, thereby providing an estimate of the uncertainty in the model's predictions. Note that the input $\mathbf{x}$ remains the same for every forward pass, meaning the variability in the predictions $\hat{\mathbf{y}}^{(t)}$ is solely due to the variability in (zeroing out) the model parameters. Therefore, this process captures the model uncertainty.

3.4 The Bayesian Denoising Autoencoder

This paper introduces a novel framework termed Bayesian Denoising Autoencoders (BDAEs), which integrates Monte Carlo dropout into denoising autoencoders (DAEs). BDAEs extend traditional DAEs by enabling the quantification of model uncertainty associated with the generated denoised images. For image data, every generated denoised image comprises multiple pixel predictions, each with its own uncertainty. BDAEs estimate the model uncertainty for every pixel, plotting a second image that represents uncertainty. Note that by integrating dropout in the autoencoder, we inherently regularize the model as well.

BDAEs maintain the architecture of traditional DAEs, but introduce stochasticity through the incorporation of MC dropout, which facilitates the estimation of uncertainty. In this paper, dropout is only implemented in the decoder network of the autoencoder. This is because the uncertainty is primarily relevant in the generation process, where the model makes predictions about the pixel values of the reconstructed image. Therefore, introducing stochasticity into the decoder allows us to capture uncertainty in the generation process, which is crucial for accurately estimating uncertainty in the output. An illustration of the BDAE is shown in Figure 3.

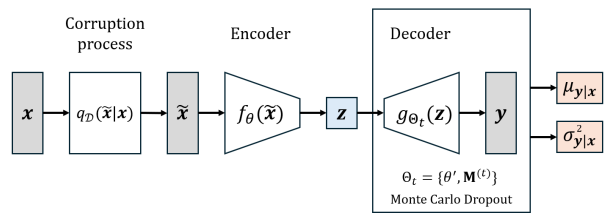

Figure 3: Architecture of the Bayesian denoising autoencoder. Monte Carlo dropout is only applied to the decoder network. The predicted image and its uncertainty are determined by averaging pixel values and calculating variance across multiple stochastic forward passes, respectively.

4 IMPLEMENTATION

Our objective is to evaluate the performance of the BDAE compared to the traditional DAE, focusing on the accuracy and their uncertainty estimates. This section covers the implementation of our methods, beginning with an introduction to the dataset used in this study. We then outline our approach to evaluating model performance, followed by a description of the network architecture. Finally, we present the training specifics.

4.1 Data

In this paper, we evaluate the performance of the BDAE on the MRI Alzheimer dataset. This dataset comprises MRI brain scan images primarily intended for Alzheimer's disease classification based on MRI scans. It is available on Kaggle and is collected from several websites, hospitals and public repositories. The images are centered and size-normalized in fixed-size grayscale images of 28×28 pixels. The dataset contains a total of 6,400 images, partitioned into training, validation, and test sets with a split of 4480/960/960 images, respectively.

4.2 Evaluation Metrics

Two different evaluation metrics are used to assess the performance of our models. These together provide a balanced evaluation of the denoising autoencoder models, assessing both the accuracy of pixel-wise predictions and the perceived quality of the images.

Mean Squared Error (MSE) is a simple and straightforward way of evaluating the performance of images [Cui et al. 2015]. It computes the squared difference between the predicted image pixel values ($\hat{\mathbf{y}}$) and actual image pixel values

($\mathbf{x}$). Then, it averages these squared differences across all pixels. Let $\mathbf{x}$ denote the true target values (i.e., vectorized image pixels) and $\hat{\mathbf{y}}$ denote the predicted values. Suppose the image has width W and height H, resulting in a total of $W \times H$ pixels. For each pixel (i, j), where $i = 1, \ldots, W$ and $j = 1, \ldots, H$), the mean squared error of a single image is calculated as

$$\text{MSE} = \frac{1}{W} \sum_{i=1}^{W} \sum_{j=1}^{H} (\hat{y}_{ij} - x_{ij})^2. \tag{11}$$

To evaluate the model's performance, we calculate the MSE for each image prediction and then average these values across the entire dataset. Note that in this paper only square images are considered, hence $W = H$.

Structural Similarity Index Measure (SSIM) is another key metric for evaluating the performance of denoising models [Brunet et al. 2011]. Unlike MSE, SSIM takes into account the perceived changes in important factors in human perception of image quality, such as structural information, luminance, and contrast of the predicted image compared to the original image. SSIM is computed using the mean values, standard deviations, and covariance of the original and predicted images. The SSIM index between ground truth image $\mathbf{x}$ and predicted image $\hat{\mathbf{y}}$ is calculated as

$$\text{SSIM}(\mathbf{x}, \hat{\mathbf{y}}) = \frac{(2\mu_{\mathbf{x}}\mu_{\hat{\mathbf{y}}} + C_1)(2\sigma_{\mathbf{x}\hat{\mathbf{y}}} + C_2)}{(\mu_{\mathbf{x}}^2 + \mu_{\hat{\mathbf{y}}}^2 + C_1)(\sigma_{\mathbf{x}}^2 + \sigma_{\hat{\mathbf{y}}}^2 + C_2)}, \tag{12}$$

where $\mu_{\mathbf{x}}, \mu_{\hat{\mathbf{y}}}$ are the means, $\sigma_{\mathbf{x}}, \sigma_{\hat{\mathbf{y}}}$ are the standard deviations, and $\sigma_{\mathbf{x}\hat{\mathbf{y}}}$ is the cross-covariance of $\mathbf{x}$ and $\hat{\mathbf{y}}$. The constants C_1 and C_2 are small constants to avoid instability when the denominator is close to zero, and are usually set to 0.01 and 0.03, respectively. This outputs a value ranging from -1 to 1. A value of 1 indicates perfect similarity between the compared images, 0 indicates no similarity and -1 indicates perfect anti-correlation, meaning the images are completely dissimilar in terms of their pixel intensity relationships. To evaluate the model's performance across an entire dataset, we average the SSIM values for all images.

4.3 Network Architecture

In this paper, we use convolutional neural networks as architectures in the (Bayesian) denoising autoencoders. It uses 3 convolutional layers in the encoder and 3 transposed convolutional layers in the decoder. We use sigmoid functions in the output layer to map the output values between 0 and 1, while applying ReLU activations to the other layers. For the Bayesian DAE, dropout layers with a consistent dropout rate of 0.1 are applied after every layer in the decoder, following the recommendations of Gal and Ghahramani [2016]. Detailed information about the neural network architectures for the models can be found in Appendix A.

4.4 Training Details

For the corruption process, we apply a noise factor of $\sigma = 0.3$. The autoencoders are trained to minimize the mean squared error as loss function. We use Adam optimizer with a batch size of 64. Additionally, we apply early stopping with a patience of 5 epochs and a maximum of 100 epochs. The models undergo hyperparameter tuning based on the learning rate through grid search. For the BDAE, we perform $T = 50$ forward passes during inference to average predictions and estimate model uncertainty, as Gal and Ghahramani [2016] have demonstrated this number to be reasonable for regression. For further details on the final model hyperparameters, we refer to Appendix B. We implement our models in PyTorch 2.3.0 and use a cloud server with an NVIDIA T4 Tensor Core GPU for training.

5 RESULTS

In this section, the results of our experiments are presented. First, we discuss the quantitative results, comparing both predictions of the BDAE and traditional DAE. Then, we evaluate the uncertainty estimates made by the BDAE by visual inspection.

5.1 Predictive Accuracy

First, we evaluate the performance of the BDAE by comparing it to the traditional DAE. These results are given in Table 1. As can be seen from the table, the DAEs outperform the BDAE in both MSE and SSIM. This suggests that the added regularization effect of incorporating Monte Carlo dropout in the BDAE may have led to over-regularization, thereby losing predictive accuracy. Thus, while BDAEs provide uncertainty estimates alongside predictions, this comes at the expense of model bias and reduced overall accuracy.

Table 1: MSE, SSIM, and the correlation between model-estimated uncertainty and actual MSE for denoising models on MRI Alzheimer test datasets. Values represent the mean and standard deviation from experiments conducted five times. Best performer is highlighted in bold.

Model	MSE $\times 10^{-3}$	SSIM	Corr.
DAE	**5.10 ± 0.13**	**0.907 ± 0.003**	—
BDAE	5.52 ± 0.20	0.089 ± 0.002	0.32 ± 0.02

Furthermore, in Figure 1, we illustrate specific predictions made by both the DAE and BDAE models, as well as the uncertainty associated with the BDAE's predictions. It can be observed that the difference between the DAE's and BDAE's predictions seems minimal.

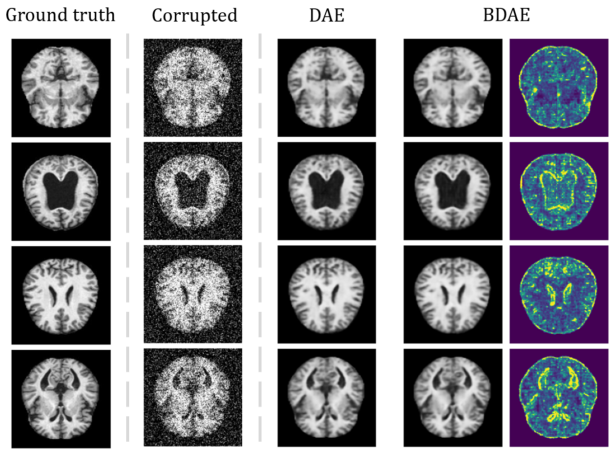

Figure 4: Out-of-sample results of DAE and BDAE applied to MRI Alzheimer dataset. Rightmost column represents uncertainty estimates of the BDAE, which are plotted on the same scale, with lighter colors indicating greater uncertainty.

5.2 Uncertainty Estimates

We evaluate the uncertainty estimates by visual inspection using the examples in Figure 1. The predictions show the most uncertainty in areas with high contrast in brightness. This is due to slight variations in intensity and pixel values at these edges, making it difficult for autoencoders to accurately reconstruct the exact edge details. Additionally, during the denoising process, fine details at the edges might be smoothed out or lost due to the nature of the DAEs reconstruction objective. Additionally, Table 1 shows us a moderate positive correlation between the uncertainty measures estimated by the model and the actual MSE, indicating that more uncertain predictions tend to have higher prediction errors.

5.3 Discussion

The results of our experiments reveal several key insights. Visual inspection shows that the Bayesian denoising autoencoder can reasonably quantify the uncertainty, but this comes at the expense of losing denoising accuracy. While machine learning models lack an inherent ground truth for model uncertainty and we cannot conduct a quantitative analysis, we do observe a positive correlation between model-estimated uncertainty and actual prediction errors. This suggests that predictions associated with higher uncertainty tend to correspond to higher prediction errors.

In this paper, the models have only been evaluated for clean datasets, with artificially added noise. In applications with medical images with natural noise, we do not know the underlying noise process. Therefore, evaluating the models on datasets with natural noise is necessary to understand their performance in real-world scenarios, which is a subject for future research.

Another limitation arises from integrating Monte Carlo dropout into the model, which inherently imposes regularization and may increase bias. Additionally, we observe that models that are not capable enough to capture certain spatial features (such as the linear models in Section **??**), indicate lower uncertainty in these regions due to their consistent failure to capture these features. For future research, we recommend exploring deep ensembles for uncertainty estimation, as well as other state-of-the-art models such as GANs or diffusion models, as they could potentially address this issue.

6 CONCLUSION

In this paper, we researched the integration of MC Dropout with DAEs to quantify model uncertainty in medical image denoising tasks, referred to as the Bayesian Denoising Autoencoder (BDAE). Our main objective was to develop a framework that preserves image quality, while also providing a quantifiable measure of uncertainty in its (denoised) image predictions.

The main takeaway from our research is that the BDAE can provide reasonable estimates of model uncertainty in generating denoised images and that these estimates are positively correlated with prediction error. However, this capability comes at the cost of predictive accuracy, resulting in an inherent trade-off between uncertainty quantification and denoising performance.

The contribution of the BDAE lies in its potential to improve decision-making in fields where image denoising is crucial, such as in medical imaging. By also providing a measure of uncertainty, BDAEs can help medical practitioners make more informed and safer decisions. However, this comes at the cost of predictive accuracy, and this should be considered in practical applications.

Further research could focus on optimizing this trade-off between uncertainty quantification and predictive accuracy. This can be addressed by integrating advanced image denoising models, such as GANs or diffusion models, which enhance predictive accuracy. Additionally, using different uncertainty estimation techniques, such as deep ensembles, can provide robust uncertainty estimates without the need for regularization as in MC dropout. Optimizing this trade-off makes incorporating uncertainty in images more useful across a wider range of applications.

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A NETWORK ARCHITECTURE

Table 2: Overview of DAE architecture.

Layer	Features in	Features out	Kernel
Conv2d	1	64	3×3 (stride=2, padding=1)
Conv2d	64	128	3×3 (stride=2, padding=1)
Conv2d	128	256	7×7
ConvTranspose2d	256	128	7×7
ConvTranspose2d	128	64	3×3 (stride=2, padding=1)
ConvTranspose2d	64	1	3×3 (stride=2, padding=1)

The (Bayesian) denoising autoencoder uses 3 convolutional layers in the encoder and 3 transposed convolutional layers in the decoder. All autoencoders use ReLU activation functions, and sigmoid for the last layer in the decoder. Detailed model architectures, including PyTorch layer names for the linear and convolutional models, are provided in Table 2, respectively. In Bayesian DAEs, where MC Dropout is implemented, we apply a dropout layer after every layer in the decoder in both DAEs with a dropout rate equal to 0.1.

B MODEL HYPERPARAMETERS

The models are trained minimizing the mean squared error as loss function, and Adam optimizer is used with a batch size of 64. Furthermore, we apply early stopping with a patience of 5 epochs and limit training to a maximum of 100 epochs. For the BDAE, we set the dropout rate equal to 0.1. We tune learning rate by testing 7 values logarithmically spaced between 10^{-2} and 10^{-4} on the validation set. The specific learning rates chosen for the final models are detailed in Table 3.

Table 3: Selected learning rates for the final denoising model configurations.

Dataset	DAE	BDAE
MRI Alzheimer	1.0×10^{-3}	1.0×10^{-3}